Highlights

Designing Multi-Compartment Last-Mile Vehicle Fleets: An Open-Source Matheuristic

- We study a fleet size and mix problem with heterogeneous multi-compartment vehicles.
- We propose a novel matheuristic approach based on a cluster-first, fleet design-second scheme.
- We present FleetMix, a spec-driven, open-source implementation of our matheuristic.
- We show the effectiveness of FleetMix compared to existing solution approaches.
- We derive several managerial insights for a real-world case study.

Designing Multi-Compartment Last-Mile Vehicle Fleets: An Open-Source Matheuristic

Abstract

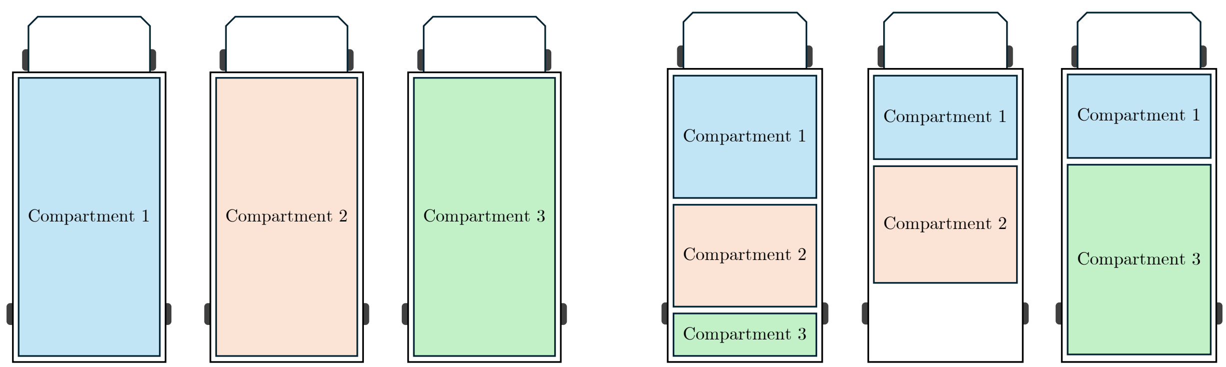
Urban food distributors are increasingly relying on flexible, multi-compartment vehicle fleets to handle diverse product mixes and challenging urban environments. While recent research on multi-compartment vehicle routing problems has explored a range of heuristics and constraints, to our knowledge, none has directly addressed fleet size and mix optimization for heterogeneous vehicles with flexible compartments in last-mile food distribution systems. Motivated by our collaboration with a major food distributor operating in Colombia, we close this practical and methodological gap by proposing a novel cluster-first, fleet design-second matheuristic approach that quickly produces cost-effective fleet size and mix solutions. To support practitioners and researchers in applying and replicating our results, we make our matheuristic publicly available through FleetMix, an open-source Python package freely available under the MIT license and distributed via PyPI. We show the effectiveness of FleetMix in finding high-quality solutions compared to existing solution approaches on available benchmark instances. Based on a real-world case study from our industry partner, which serves hundreds of customers daily, we also show the value of utilizing multi-compartment vehicles to reduce fleet sizes and total logistics costs.

Keywords: fleet design, multi-compartment, matheuristic, open-source Python package, last mile

1. Introduction

The growing expansion of the food industry has drawn increasing attention to food safety and quality management during production, storage, and transportation (Hsiao et al., 2017). Food service companies and grocery stores typically receive a range of product categories from distributors, each requiring specific temperature conditions defined by legal regulations or quality standards (Hertog et al., 2014; Ostermeier and Hübner, 2018). Consequently, food distributors must be able to transport multiple temperature-sensitive product types simultaneously. However, due to their distinct environmental needs, these products cannot be stored together in the same space during transit (Chen et al., 2019). This makes planning efficient delivery operations both increasingly challenging and essential, as delivery costs account for 20% of overall food supply chain costs (Kuhn and Sternbeck, 2013).

To address this challenge, distributors have increasingly turned to fleets of Multi-Compartment Vehicles (MCVs) in recent years, which enable the simultaneous transport of multiple product categories (Ostermeier et al., 2021). MCVs are generally designed with either fixed or flexible compartments. MCVs with fixed compartments have predefined capacities determined during manufacturing and cannot be altered, whereas MCVs with flexible compartments offer the ability to adjust their sizes (Henke et al., 2015). In the context of food distribution, MCVs commonly have three compartments for dry, refrigerated, and frozen foods, with adjustable sizes to match load requirements. Figure 1a illustrates three Single-Compartment Vehicles (SCVs), each dedicated to transporting a single product type. In contrast, Figure 1b shows three possible configurations of the same flexible-compartment MCV, illustrating that the number and size of compartments can be adjusted according to the combination and volume of product categories to be transported. A real-world example of MCVs is provided by Delivery Concepts, whose Body Series vehicles integrate insulated compartments that can be configured for frozen, refrigerated, heated, and ambient products within a single truck (Delivery Concepts, 2026). Companies like FreshDirect, Sysco, and Whole Foods Markets have reported the benefits of using MCVs, including improved customer experience and reduced fuel costs, emissions, traffic exposure, and parking issues (He et al., 2024). By consolidating different product categories, MCVs also reduces the number of vehicles needed to meet demand compared to using separate SCVs.



(a) Three SCVs carrying a single product type each.

(b) Three possible configurations of a single MCV.

Figure 1: Comparison between SCVs and a flexible-compartment MCV.

Motivated by our collaboration with a major food distributor operating in Bogotá, Colombia, we investigate the problem of designing MCV fleets for last-mile food distribution systems. This problem considers several important real-world features of modern delivery operations with MCVs, including heterogeneous vehicles with flexible compartments, both fixed and variable vehicle usage costs, route duration constraints, and a large number of customers. The problem consists of determining three interconnected decisions: (i) the number of each vehicle type to use, (ii) the type, number, and size of compartments assigned to each vehicle, and (iii) the distribution plan (i.e., demand allocation) for each vehicle. The objective is to minimize total fixed and variable vehicle costs while satisfying customer demand for different product types.

A review of the existing literature on MCV-based distribution systems reveals several limitations, which we discuss in further detail in Section 2. Most studies focus on routing decisions with fixed vehicle compartment configurations, which restricts the ability to adjust vehicle capacity to customer demand (Ostermeier et al., 2021). While some works consider flexible compartments, most of them do not incorporate decisions on fleet size and mix and are limited to operational routing decisions. Furthermore, closely related studies on fleet size and mix with MCVs rely on homogeneous fleets and do not account for practical constraints, such as maximum route durations. These limitations are particularly relevant in real-world food distribution settings, including those faced by our partner company, where multiple vehicle types are available and drivers are subject to maximum working hours imposed by labor regulations.

In response to these research gaps, our main methodological and practical contributions can be summarized as follows.

1. **Novel Matheuristic Approach.** We propose a matheuristic approach that addresses practical fleet size and mix problems with heterogeneous MCVs under load capacity and maximum route duration constraints. An appealing property of our matheuristic is that it can be easily adapted to existing variants addressed in the literature (see Section 2), including homogeneous vehicles, fixed compartments, and single- and multi-stop policies (where customers are allowed to be served via a single or multiple vehicle stops, respectively). This makes our matheuristic a unified and versatile approach suitable for a wide range of applications.

2. **Open-Source Implementation.** To support practitioners and decision-makers in applying and replicating our findings, we introduce FleetMix, an open-source Python package implementing our matheuristic, available on GitHub (`https://github.com/ekohan/fleetmix`) and PyPI under the MIT license. To the best of our knowledge, FleetMix is the first open-source implementation of a solution approach for this problem class. FleetMix features a traceable and modular architecture with interchangeable components, allowing for easy extension. To facilitate adoption by diverse users, we develop three interfaces: a command-line tool, a Python API, and a web-based graphical interface for fleet managers.

3. **Numerical Validation and Managerial Insights.** We test our matheuristic approach using both synthetic instances from the literature and a real-world dataset from a food distributor operating in Bogotá, Colombia, which serves hundreds of customers daily. Our results show the effectiveness of our matheuristic approach compared to benchmark solution approaches, generating high-quality fleet designs in seconds. For real-world instances, we illustrate the benefits of using MCVs to reduce both total costs and the number of vehicles compared to traditional SCV fleets.

The rest of the manuscript is structured as follows. We review the relevant literature in Section 2, highlighting the research gaps we address and the main contributions of our work. Section 3 defines the problem under investigation. In Section 4, we describe our matheuristic approach, and Section 5 presents FleetMix, our open-source Python package. Section 6 shows the effectiveness of our matheuristic approach over synthetic instances from the literature, while Section 7 reports the results and managerial insights obtained using the real-world case study. Lastly, Section 8 outlines our conclusions and describe several research avenues for future research.

2. Literature Review

In this section, we review three fields of work related to MCVs. We begin by analyzing the literature on Multi-Compartment Vehicle Routing Problems (MCVRPs). We then review the literature related to fleet composition problems. We conclude this section by describing the main research gaps addressed in our work.

2.1. Multi-Compartment Vehicle Routing Problems

The MCVRP is one of the most important Vehicle Routing Problem (VRP) variants that considers that vehicles' load capacity can be divided into different compartments. A recent comprehensive review of MCVRPs is presented by Ostermeier et al. (2021), who propose a taxonomy based on attributes related to compartment characteristics and order fulfillment. As highlighted by Ostermeier et al. (2021), MCVRPs find many practical applications, including fuel distribution, waste collection, agricultural supply chains, maritime transportation, and food distribution. However, most literature focused on MCVs with fixed compartments, meaning that the number and capacity of compartments cannot be adjusted to match load requirements. For instance, Mendoza et al. (2010) address a MCVRP with stochastic demands, assuming fixed compartment sizes and an unlimited fleet of homogeneous vehicles. The authors formulate the problem as a two-stage stochastic programming model and develop a memetic algorithm with solution evaluation and reparation strategies that account for the stochastic nature of the problem. Mendoza et al. (2010) show the effectiveness of their mementic algorithm using synthetic instances with up to 480 customers. Later, Reed et al. (2014) study a MCVRP in the context of household solid waste recycling collection. They consider a homogeneous fleet with two fixed compartments and propose an ant colony heuristic approach. The authors test their approach on synthetic instances with 50 customers and two types of products.

To the best of our knowledge, Henke et al. (2015) are the first to introduce the MCVRP with flexible compartment sizes in the context of glass waste collection. Assuming a predefined, homogeneous fleet of vehicles (i.e., no fleet sizing decisions are considered), the authors formulate a Mixed-Integer Linear Programming (MILP) model to minimize the total distance traveled by the vehicles. Henke et al. (2015) also devise a variable neighborhood search algorithm that allows visiting infeasible solutions in terms of vehicle capacity. Their experiments show that their MILP model can solve instances with up to 10 customers and apply their heuristic to instances with up to 50 customers. Later, Henke et al. (2019) propose a branch-and-cut algorithm to address the problem they had previously introduced in Henke et al. (2015). We refer the reader to Ostermeier et al. (2021) for other related works focusing on routing MCVs.

Now, in the context of food distribution, Hübner and Ostermeier (2019) investigate the impact of loading and unloading costs when routing a homogeneous fleet of MCVs with flexible compartments.

The authors propose a large neighborhood search heuristic with tailored removal and insertion operations. Their experiments show that MCVs can reduce total costs by up to 6.3% compared to a SCV fleet. Another study that considers two-dimensional loading constraints can be found in Ostermeier et al. (2018). Next, Chen et al. (2019) study a MCVRP with fixed compartments and time windows. Their Adaptive Large Neighborhood Search (ALNS) heuristic is applied to a real-world case study from a cold-chain distribution company in Shanghai, showing that total costs increase as time windows become tighter, mainly due to higher fuel consumption. Concurrently, Martins et al. (2019) consider a MCVRP with flexible compartments and product-oriented time-window constraints. The authors also propose an ALNS heuristic, which is tested using instances from Kovacs et al. (2014). Martins et al. (2019) show that scheduling time windows in advance, rather than assigning them after the daily plans are made, leads to more on-time deliveries. Later, Heßler (2021) proposes exact algorithms for the MCVRP with flexible compartments. In contrast to Henke et al. (2015), who assume compartment sizes are restricted to predefined, equally sized positions, they also consider a continuous setting where compartment capacities can vary freely, as in our case. Their results show that these exact approaches can solve instances with up to 50 customers to optimality. Frank et al. (2021) address a periodic MCVRP that includes the definition of store-specific delivery patterns and decisions on delivery schedules. The authors develop a metaheuristic in which delivery patterns are determined via ALNS, while a Large Neighborhood Search (LNS) component is used to define vehicle routes within each period. Another extension of the MCVRP is examined in Wang et al. (2023), who consider electric vehicles, time windows, and temperature settings. The interested reader can refer to the recent literature review presented by Guerriero et al. (2025) for other works addressing VRPs with perishable products.

2.2. Fleet Size and Mix Problems

As can be observed in the previous section, most literature on MCVs focuses on routing a predefined and fixed fleet of homogeneous vehicles, without considering decisions about fleet size and mix. While fleet size and mix problems have recently attracted growing attention due to their practical relevance, most literature concentrates on distribution problems involving SCVs (Hoff et al., 2010). Specifically, since the seminal work of Golden et al. (1984), who introduce the fleet size and mix problem considering SCVs with fixed acquisition and variable routing costs, the literature has progressively incorporated various real-world features, such as delivery time windows (Liu and Shen, 1999; Bräysy et al., 2008; Repoussis and Tarantilis, 2010), limited number of vehicles (Koç et al., 2016), electric vehicles (Hiermann et al., 2016; González-Rodríguez et al., 2024), and stochastic customer demand (Loxton et al., 2012; Zinnenlauf et al., 2024), among others. However, to the best of our knowledge, the only contribution addressing fleet size and mix decisions with MCVs has been proposed by Ostermeier and Hübner (2018).

Ostermeier and Hübner (2018) examine a MCVRP with vehicle selection decisions. However, the authors only consider two homogeneous vehicle types: SCVs and MCVs, meaning that all vehicles within each type have identical characteristics (e.g., the same maximum load capacity). Ostermeier and Hübner (2018) propose a mathematical programming formulation aiming to find the number and routes for SCVs and MCVs that minimizes a non-linear cost function composed by fixed cost per active compartment, transportation costs based on distance traveled, and a fixed cost per vehicle stop. They also propose a LNS heuristic based on their previous work in Hübner and Ostermeier (2019). Using a real-world case study from a grocery retailer, Ostermeier and Hübner (2018) show that using a mixed fleet of SCVs and MCVs can reduce total costs by up to 30%.

2.3. Summary of Main Research Gaps

Our literature review identifies several practical and methodological research gaps. As highlighted in the comprehensive review presented by Ostermeier and Hübner (2018), most existing studies have focused on routing a homogeneous fleet of MCVs with fixed compartments, implying that vehicle capacities cannot be adapted to better match customer demand. Table 1 highlights, to the best of our knowledge, the works most closely related to ours that address problems with flexible compartments. Except for Ostermeier and Hübner (2018), the scope of existing works is limited to operational decisions related to vehicle routing, without considering decisions on fleet size and mix. Note that none of the studies listed in Table 1 provide open-source implementations, which impedes reproducibility and limits comparative evaluation.

Table 1: Synthesis of related works with flexible compartments.

Work	Problem attributes			Decisions		
	Compartment size	Vehicle fleet	SCV-MCV mix	Fleet size	Fleet mix	Routing
Henke et al. (2015)	Discrete	Homogeneous	MCV	×	×	Explicit
Henke et al. (2019)	Discrete	Homogeneous	MCV	×	×	Explicit
Hübner and Ostermeier (2019)	Continuous	Homogeneous	MCV	×	×	Explicit
Martins et al. (2019)	Continuous	Homogeneous	MCV	×	×	Explicit
Heßler (2021)	Continuous	Homogeneous	MCV	×	×	Explicit
Frank et al. (2021)	Continuous	Homogeneous	MCV	×	×	Explicit
Wang et al. (2023)	Continuous	Homogeneous	MCV	×	×	Explicit
Ostermeier and Hübner (2018)	Continuous	Homogeneous	Both	✓	✓	Explicit
This work	**Continuous**	**Heterogeneous**	**Both**	✓	✓	**Explicit & Implicit**

The work most closely related to ours is that of Ostermeier and Hübner (2018). However, Ostermeier and Hübner (2018) assume a homogeneous vehicle fleet consisting of a single SCV type and a single MCV type, which does not reflect the context of our partner company and other food distributors, where multiple vehicle types are available, such as conventional trucks and small delivery vans. Similar to Ostermeier and Hübner (2018), we consider a fixed cost associated with dispatching and loading a vehicle with a specific compartment configuration, as well as a variable transportation cost. However, in contrast to Ostermeier and Hübner (2018), where routing decisions are modeled explicitly within their mixed-integer non-linear program and LNS procedure, we use routing information to estimate variable transportation costs and verify compliance with maximum route duration constraints. In our matheuristic, this routing information can be obtained either *implicitly* through continuous approximation, which is computationally efficient for large-scale instances, or *explicitly* using a state-of-the-art routing solver, which additionally enables the construction of the corresponding vehicle routes during our customer cluster generation phase (see Section 4.2 for further details). The inclusion of maximum route duration constraints, which are not considered in Ostermeier and Hübner (2018), is motivated by the operational practices of our partner company and by labor regulations that limit drivers' working time during a workday.

As discussed in Section 1 in more detail, we address these gaps by investigating a comprehensive variant of the fleet size and mix problem with MCVs. This problem considers heterogeneous vehicles, flexible compartments, fixed vehicle and compartment costs, variable transportation costs, single- and multi-stop delivery policies, and constraints on vehicle capacity and route duration. To solve realistic instances, we propose the first open-source solution approach, FleetMix, and show its effectiveness on both available synthetic benchmark instances from the literature and a real-world case study using data from our partner company, a food distributor in Bogotá, Colombia.

3. Problem Definition

We now formally introduce the problem under investigation. We consider a single depot with sufficient inventory at the beginning of the delivery horizon to satisfy all customer demands. Let $N' = \{0, 1, ..., n\}$ be the set of nodes, where 0 denotes the depot and $N = \{1, ..., n\}$ the customers. Further, let P be the set of product types (e.g., dry, chilled, and frozen) and q_p the units of vehicle load capacity required to transport one unit of product type $p \in P$. Each customer $i \in N$ has a demand of d_{ip} units of product type $p \in P$. We consider a multi-stop policy in which customers may receive deliveries from multiple vehicles, but each product type must still be delivered in full by one vehicle, as splitting is not practical (Gu et al., 2024).

The delivery fleet comprises a set of heterogeneous vehicle types, each defined by specific operational characteristics such as maximum load capacity, average speed, and fixed and variable costs. Each vehicle type can divide its load capacity into a set M of flexible compartments, each designed for a specific product type. In other words, vehicles can be configured in multiple ways by assigning different subsets of compartments. To capture this flexibility, we introduce the concept of *vehicle configuration*, defined as a certain vehicle type equipped with a (non-empty) subset of compartments. The set of all vehicle configurations is denoted by V. For example, if the company operates two vehicle types and there are three available compartments, each type can be configured in $2^3 - 1 = 7$ distinct ways, resulting in $2 \times 7 = 14$ configurations in V. Based on the corresponding vehicle type, each vehicle configuration $v \in V$ has a maximum load capacity of Q_v, which can be distributed among its compartments, and a maximum route time duration of T_v.

Given our interest in addressing practical food distribution problems involving hundreds of customers, we aggregate customers into multiple capacitated clusters, denoted by the set K. Since these clusters are not necessarily mutually exclusive subsets of N, we define $K_i \subseteq K$ as the subset of clusters containing customer $i \in N$. As we explain in further detail in Section 4, clusters are created based on the set of vehicle configurations V. Notably, we define $K_v \subseteq K$ as the subset of clusters that can be feasibly served by vehicle configuration $v \in V$ regarding maximum load capacity (Q_v) and route time duration (T_v) constraints. Similarly, let $V_k \subseteq V$ be the set of vehicle configurations that can feasibly serve cluster $k \in K$. The total cost of serving cluster $k \in K_v$ with vehicle configuration $v \in V$, denoted by c_{vk}, consists of a fixed cost incurring for dispatching the vehicle and equipping it with the set of compartments associated with the configuration, along with a variable cost based on the distance traveled to serve all customers in the cluster. Table 2 lists our mathematical notation, including the notation later presented in Section 4.

The problem involves determining the optimal MCV fleet size and mix by deciding the number of each vehicle configuration to use. Specifically, this includes selecting the appropriate vehicle types, defining their compartmentalization (i.e., number, type, and load capacity associated with each compartment), and assigning each selected vehicle configuration to a specific customer cluster. The objective is to minimize the total fixed and variable vehicle costs while meeting the individual demand of each customer for different product types. Note that, as we address a tactical fleet size and mix problem, operational routing decisions are not explicitly modeled but are implicitly accounted for when verifying that each customer cluster adheres to maximum route time constraints associated with the respective vehicle configuration.

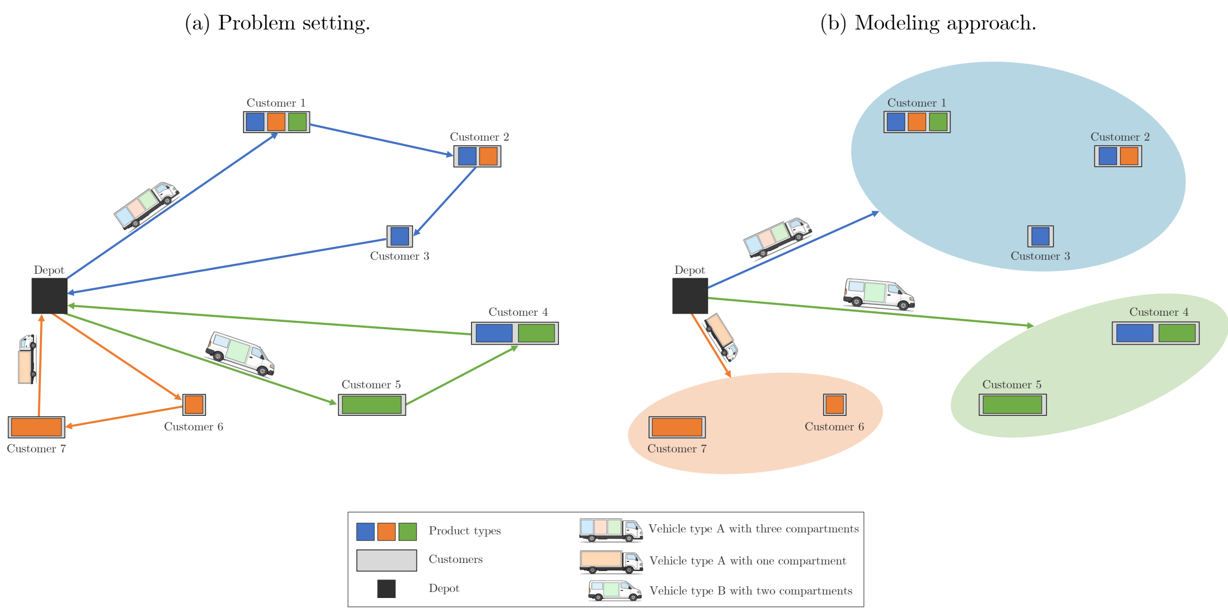


Figure 2: Illustrative example of the problem setting and modeling approach.

Figure 2 shows an example of a problem instance with seven customers. Each customer demands between one and three different product types. Two vehicle types are available: Vehicle Type A, representing a small delivery van, and Vehicle Type B, representing a conventional truck. Figure 2(a) shows a feasible solution in which three vehicle configurations are selected: one van with two compartments and two trucks, one equipped with three compartments and the other with a single compartment. The truck with three compartments delivers three product types to Customer 1, two to

Customer 2, and one to Customer 3. The van serves Customers 4 and 5, while the single-compartment truck delivers one product type to Customers 6 and 7. Figure 2(b) illustrates our modeling approach based on clusters, where each selected vehicle configuration is assigned to a particular cluster.

Table 2: Mathematical notation.

Symbol	Description
Sets	
N'	Set of nodes (including depot 0)
N	Set of customers, i.e., $N = N' \setminus \{0\}$
P	Set of product types
M	Set of compartment types
V	Set of vehicle configurations
K	Set of customer clusters
$K_i \subseteq K$	Set of clusters containing customer $i \in N$
$K_v \subseteq K$	Set of clusters feasible for configuration $v \in V$
$V_k \subseteq V$	Set of configurations that can serve cluster $k \in K$
$P_i \subseteq P$	Set of product types demanded by customer $i \in N$
S_i^p	Set of subsets of P_i that contain product $p \in P_i$
K_i^s	Set of clusters that can serve subset $s \in S_i^p$ for customer $i \in N$
Parameters	
d_i^p	Demand of customer $i \in N$ for product $p \in P$
q_p	Capacity required per unit of product $p \in P$
Q_v	Maximum load capacity of configuration $v \in V$
T_v	Maximum route duration of configuration $v \in V$
c_{vk}	Cost of serving cluster $k \in K_v$ with configuration $v \in V$
t_{vk}	Estimated route duration for cluster $k \in K$ with configuration $v \in V$
δ_{vk}	Line-haul travel time between depot and cluster $k \in K$
β	Constant in BHH routing approximation
γ	Service time per customer
A_k	Service area size of cluster $k \in K$
λ	Weight parameter in composite distance metric
D_{ij}^{geo}	Geographical distance between customers $i, j \in N$
D_{ij}^{prod}	Product-demand similarity distance between customers $i, j \in N$
D_{ij}	Composite distance metric between customers $i, j \in N$
η	Minimum cluster size
Δ	Number of neighboring clusters considered for merging
Decision variables	
x_{vk}	1, if configuration $v \in V$ serves cluster $k \in K_v$. 0, otherwise.

4. Matheuristic Approach

Given our interest in designing MCV fleets for practical food distribution operations, we propose a novel matheuristic approach based on a cluster-first, fleet design-second scheme (see Archetti and

Speranza (2014) for a survey on matheuristics applied to VRPs). Figure 3 presents a high-level overview of the main steps involved. First, we define the set of possible vehicle configurations based on the available vehicle types and compartments. Subsequently, for each vehicle configuration, we generate a set of feasible customer clusters using tailored clustering techniques. We then determine the optimal fleet size and mix through an integer programming model. To further enhance the solution, we apply an improvement phase to create and include additional feasible customer clusters. This procedure repeats until no further improvements can be reached, at which point we return the best fleet size and mix found.

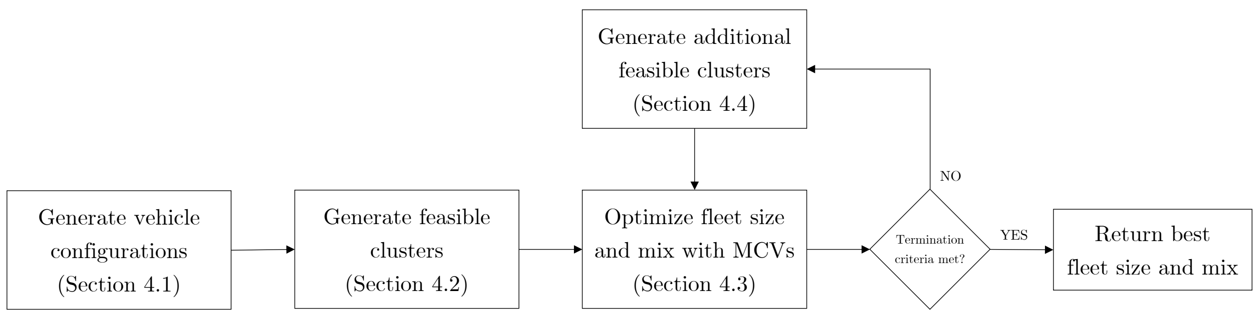


Figure 3: High-level overview of the matheuristic approach.

4.1. Generate Vehicle Configurations

We define the set of vehicle configurations, V, based on the available vehicle types and compartments. Specifically, for each vehicle type, we define $2^{|M|} - 1$ configurations, where M denotes the set of compartments. For example, the food distributor that inspired this research divides their vehicles into three compartments dedicated to dry (m^{dry}), refrigerated (m^{ref}), and frozen (m^{fro}) product types. In this case, we consider the following subsets of M, each defining a configuration: $\{m^{\text{dry}}\}$, $\{m^{\text{ref}}\}$, $\{m^{\text{fro}}\}$, $\{m^{\text{dry}}, m^{\text{ref}}\}$, $\{m^{\text{dry}}, m^{\text{fro}}\}$, $\{m^{\text{ref}}, m^{\text{fro}}\}$, and $\{m^{\text{dry}}, m^{\text{ref}}, m^{\text{fro}}\}$. Therefore, we generate $r \cdot (2^{|M|} - 1)$ vehicle configurations, with r denoting the number of vehicle types.

Note that our procedure includes generating SCVs dedicated to a particular product type (namely, the configurations arising from the singletons $\{m^{\text{dry}}\}$, $\{m^{\text{ref}}\}$, and $\{m^{\text{fro}}\}$), enabling our fleet size and mix optimization model to select either MCVs or SCVs based on their respective cost structures and characteristics. Since we consider multiple heterogeneous vehicle types, our procedure generalizes the approach presented by Ostermeier and Hübner (2018), who only consider one vehicle type. Lastly, since we consider MCVs with flexible compartments, note that our optimization model then defines the specific load capacity associated with each compartment of the selected vehicle configurations.

4.2. Generate Feasible Customer Clusters

After defining the set of vehicle configurations, V, we generate feasible clusters of customers that can be served by at least one configuration while satisfying maximum load capacity and route duration constraints. Specifically, for each configuration $v \in V$, we apply a variety of clustering techniques to create clusters that are feasible for that configuration. These techniques include k-means, k-medoids, Gaussian mixture models, and hierarchical (agglomerative) clustering (see Appendix A for a summary of the clustering techniques). While these are standard clustering techniques, we adapt them to account for both geographical and demand proximity between customers. Furthermore, we apply a split-and-merge procedure to the resulting set of clusters, not only to validate feasibility but also to enrich the final set of clusters that serves as input to our optimization phase.

Note that, as described in Section 3, the resulting clusters do not necessarily partition the set of customers, as some customers may appear in more than one cluster. However, the actual allocation of each customer to a given cluster (and, therefore, vehicle configuration) is then determined during our optimization phase (see Section 4.3).

In the following, we explain how load capacity and route duration feasibility are verified, as well as how the distance metric is adapted for the clustering techniques used.

4.2.1. Load Capacity and Route Duration Feasibility Checks

To ensure the creation of feasible customer clusters, we perform both capacity and route duration feasibility checks. Capacity constraints require that the total demand across all product types within a cluster does not exceed the total capacity of the corresponding vehicle configuration, while route duration constraints ensure that the total travel time to serve a cluster remains within a prescribed threshold. Since the total capacity required by a cluster can be readily determined by summing the demand across all product types, we now focus on how route durations are computed within the context of our tactical decision-making problem. We then explain how both capacity and route duration feasibility are ensured through a split-and-merge procedure.

Computation of vehicle route durations. Depending on the application context, our matheuristic allows decision-makers to verify maximum route duration constraints using one of two methods: Continuous Approximation (CA) or a state-of-the-art routing solver. CA is a widely used technique to approximate vehicle routing costs in large-scale integrated decision-making problems where operational routing decisions play a subordinate role, such as location-routing (Winkenbach et al., 2016; Pina-Pardo et al., 2022), workforce sizing and routing (Mandal et al., 2025), and fleet design and routing problems (Franceschetti et al., 2017; Zinnenlauf et al., 2024). In this work, we consider the seminal CA formula proposed by Beardwood et al. (1959), extended to incorporate vehicle setup times, customer service times, and depot-to-cluster travel times. Specifically, for a given cluster $k \in K_v$ with n_k customers distributed over a service area A_k, we approximate the total routing time of serving the cluster using vehicle configuration $v \in V$ as

$$t_{vk} \approx 2 \cdot \delta_{vk} + \beta \cdot \sqrt{n_k \cdot A_k} + \gamma \cdot n_k, \tag{1}$$

where δ_{vk} is the line-haul travel time between the depot and the centroid of the cluster, γ the customer service time, and β is a non-negative constant. The second term corresponds to the well-known Bearwood-Halton-Hammersley (BHH) approximation presented by Beardwood et al. (1959) (see Silva et al. (2022) for a recent work showing the quality of this approximation to produce near-optimal estimates).

Note that an alternative method for computing vehicle route durations is to employ an existing routing solver. In our matheuristic implementation (see our GitHub repository), decision-makers can also use the PyVRP library developed by Wouda et al. (2024), which is recognized as a state-of-the-art vehicle routing solver. While this approach might enhance vehicle routing time estimations, it often requires substantially more computational time than the CA-based approach, especially when dealing with large instances. However, integrating PyVRP into our matheuristic also enables the explicit construction of vehicle routes within each generated customer cluster, which may be particularly useful when decision-makers need not only to determine the fleet size and mix, but also to define a detailed routing plan for the vehicles.

Algorithm 1 Recursive Cluster Splitting

```
function SplitCluster(Cluster, DepthLevel, MaxDepth)
    if ExceedsConstraints(Cluster) and DepthLevel < MaxDepth then
        SubClusters ← Split(Cluster, 2)                ▷ Split cluster into two sub-clusters
        for each SubCluster in SubClusters do
            SplitCluster(SubCluster, DepthLevel + 1, MaxDepth)
        end for
    else
        AddValidCluster(Cluster)
    end if
end function
```

Split-and-merge procedure. After computing vehicle route durations, we verify both load capacity and route duration constraints through a split-and-merge procedure. First, we conduct a recursive binary splitting, as detailed in Algorithm 1. When cluster $k \in K_v$ exceeds either the total capacity or the maximum route duration limit associated with vehicle configuration $v \in V$ (i.e., when $\sum_{i\in k}\sum_{p\in P} d_{ip} > Q_v$ or $t_{vk} > T_v$, respectively), the cluster is recursively divided into two subclusters. This splitting process continues until all resulting clusters satisfy both constraints or a predefined maximum recursion depth is reached.

Since Algorithm 1 may produce small clusters with only a few customers, we then apply a merging step to all clusters $k \in K$ such that $|k| < \eta$, where η denotes a minimum cluster size. For each such small cluster k_i, we define a neighborhood consisting of the Δ closest clusters regarding centroid-to-centroid distance. Then, for each neighboring cluster k_j, we check whether the combined cluster $k_i \cup k_j$ can be feasibly served by a single vehicle configuration, considering both load capacity and route duration constraints. If feasible, we update the set of clusters by adding $k_i \cup k_j$, i.e., $K = K \cup \{k_i \cup k_j\}$. Note that a similar merging procedure is used during our improvement phase, but only to promising clusters selected in optimal solutions (see Section 4.4).

4.2.2. Distance Metric

Lastly, we describe how we modify the distance metric to account for both geographic and demand proximity when generating clusters. To enhance the clustering process, we create a composite distance metric that combines geographical distance and product demand similarity into a single value. We thus define a composite distance function D_{ij} between customers $i, j \in N$ as

$$D_{ij} = \lambda \cdot D_{ij}^{\text{geo}} + (1 - \lambda) \cdot D_{ij}^{\text{prod}} \tag{2}$$

where $\lambda \in [0, 1]$ establishes the weight between geographical proximity, measured by the geographical distance D_{ij}^{geo}, and product demand similarity, given by the product distance D_{ij}^{prod} (we refer the reader to Appendix A for the specific value of λ used in each clustering technique considered). Notably, we compute the product distance (measuring demand similarity) as the cosine distance between weighted demand composition profiles, where demands are normalized by total volume and weighted equally across product types.

4.3. Optimize Fleet Size and Mix with Heterogeneous MCVs

Based on the generated set of clusters, we now define the MCV fleet size and mix that minimizes the total costs incurred to meet each individual customer's demand. It is worth recalling that the total cost of serving cluster $k \in K_v$ with vehicle configuration $v \in V$, denoted by c_{vk}, includes both a fixed cost (associated with setting up the vehicle's compartments and dispatching it) and a variable cost that depends on the distance traveled to serve the cluster. Notably, such fixed costs capture the additional operational effort required to set up and operate a MCV configuration, as opposed to a SCV configuration.

Let x_{vk} be a binary decision variable equal to 1 if and only if vehicle configuration $v \in V$ serves cluster $k \in K_v$. Our integer programming model reads

$$\text{minimize} \quad \sum_{v \in V} \sum_{k \in K_v} c_{vk} \cdot x_{vk}, \tag{3a}$$

subject to

$$\sum_{k \in K_i} \sum_{v \in V_k} x_{vk} = 1, \qquad i \in N, \tag{3b}$$

$$\sum_{v \in V_k} x_{vk} \leq 1, \qquad k \in K, \tag{3c}$$

$$x_{vk} \in \{0, 1\}, \qquad v \in V, k \in K_v. \tag{3d}$$

The objective function (3a) minimizes the total fixed and variable costs. Constraints (3b) force to serve each individual customer $i \in N$. Constraints (3c) ensure that each cluster is visited by at most one vehicle configuration. Lastly, Constraints (3d) define the domain of the decision variables.

Note that, after solving Model (3), the specific load capacity required for each compartment of the selected vehicle configurations can be readily determined by summing the demand for each product type across all customers in the respective cluster. Further, note that Model (3) is an exact extended formulation of our problem when all feasible customer clusters are included. However, solving this complete formulation is computationally intractable for realistic instances, as the number of feasible clusters increases exponentially with the number of customers.

Multi-stop delivery policy. Note that Constraints (3b) enforce a *single-stop policy*, where the entire demand for each product type of each customer must be fully served by a single vehicle. Let us introduce the following additional notation to accommodate a multi-stop delivery policy. Let P_i be the set of product types demanded by customer $i \in N$. We define $S_i^p = \{P' \subseteq P_i : p \in P'\}$ as the set of all subsets of P_i that contain product $p \in P_i$. Lastly, for each product type $p \in P_i$, let K_i^s be the subset of clusters containing customer $i \in N$ that can serve the demand of all product types in $s \in S_i^p$. Thus, the following constraints

$$\sum_{s \in S_i^p} \sum_{k \in K_i^s} \sum_{v \in V_k} x_{vk} \;=\; 1, \qquad i \in N, p \in P_i, \tag{3e}$$

must replace Constraints (3b) in Model (3) to allow multiple deliveries per customer.

4.4. Improvement Phase

We perform an improvement phase on the MCV fleet size and mix found by Model (3). Specifically, after solving Model (3) for a given set of clusters K, we consider each pair of clusters (k_i, k_j) selected in the optimal solution $\boldsymbol{x}^*$ (i.e., $x^*_{k_i,v_i} = 1$ and $x^*_{k_j,v_j} = 1$ for some $v_i, v_j \in V$), check if a single vehicle configuration can feasibly serve them together based on the load capacity and route duration constraints, and, if feasible, update the set of clusters by adding $k_i \cup k_j$, i.e., $K = K \cup \{k_i \cup k_j\}$. After finishing updating the set of clusters, we solve Model (3) and repeat this procedure until no further improvement can be reached. Note that, unlike the merging procedure performed during the clustering phase (see Section 4.2), which considers only small clusters that are geographically close, our improvement phase evaluates all possible pairs of clusters (selected in the optimal solution at hand) regardless of their size or spatial proximity.

5. FleetMix: A Spec-Driven Python Package

FleetMix is the executable companion to our matheuristic: a Python package that implements all algorithms and models presented in Section 4. Available on PyPI and at `https://github.com/ekohan/fleetmix` under the MIT license, FleetMix serves two audiences. Researchers can reproduce our results of Section 6 with a single command and extend the algorithms by swapping components, and practitioners can deploy our matheuristic directly in production. The version used for this paper (`v1.1.0`) is permanently archived to ensure reproducibility.

5.1. Architecture and Design

FleetMix implements our matheuristic through four core modules that mirror the paper's structure: (i) a clustering engine that generates feasible customer clusters using the techniques described in Section 4.2, (ii) an optimization core that solves the integer programming formulation presented in Section 4.3, (iii) a post-optimization refinement module implementing the improvement phase detailed in Section 4.4, and (iv) a benchmarking framework for systematic evaluation.

The implementation uses well-defined interfaces between components. Researchers can substitute alternative clustering algorithms, route time estimators, or MILP solvers without modifying other modules. This modularity is systematically documented in our specification system, which serves as the blueprint for the entire architecture.

5.2. Specification System

To address the semantic gap between mathematical description and software implementation in computational research, we connect the paper's mathematical formulation to its software through machine-readable specification documents. Each specification details a component's mathematical basis, interface contract, implementation, and usage examples. For example, `clustering.md` documents the split-and-merge procedure (Algorithm 1), its default parameters, and how edge cases are handled, providing context that is difficult to extract from source code alone. This specification layer enables reviewers to trace equations to implementations and researchers to extend the methodology with confidence.

Table 3 illustrates this traceability, mapping key elements of our matheuristic from the paper to the corresponding software counterparts.

Table 3: Traceability between paper sections, specification documents, and implementation modules.

Paper Section	**Specification Document**	**Implementation Module**
§4.1 Vehicle Configs	`vehicle_configurations.md`	`utils/vehicle_configurations.py`
§4.2 Clustering	`clustering.md`	`clustering/generator.py`
§4.3 Optimization	`optimization.md`	`optimization/core.py`
§4.4 Improvement	`post_optimization.md`	`post_optimization/merge_phase.py`

5.3. Usage

FleetMix exposes three interfaces. The command-line tool serves as the primary entry point (e.g., `fleetmix optimize --demand customers.csv --config fleet.yaml`), and the command `fleetmix reproduce-paper` regenerates every experiment in Sections 6 and 7. The Python API (`import fleetmix as fm`) enables programmatic integration. The web interface allows interactive parameter exploration without coding. All three interfaces use identical input-output schemas, ensuring reproducibility across modalities. The repository includes worked examples: `custom_clustering.py` implements a round-robin clustering method, `custom_route_time.py` provides a straight-line route time estimator, `custom_solver_adapter.py` integrates a solver with relaxed optimality gap, and `heterogeneous_fleet.py` demonstrates configuring mixed fleets with product-specific vehicle constraints. A typical usage of the Python API is illustrated in Listing 1.

Listing 1: Example usage of the FleetMix Python API.

```python
# CLI equivalent: fleetmix optimize --demand customers.csv --config fleet.yaml
# Or programmatically via the Python API:

import fleetmix as fm

solution = fm.optimize(
    demand="customers.csv",
    config="fleet_config.yaml",  # YAML file or FleetmixParams object
)

print(f"Total vehicles: {solution.total_vehicles}")
print(f"Total cost: ${solution.total_cost:,.2f}")
print(f"Vehicles by type: {solution.vehicles_used}")

for cluster in solution.selected_clusters:
    print(f"  {cluster.vehicle_type}: "
          f"{len(cluster.customers)} customers, "
          f"demand: {cluster.total_demand}")
```

6. Effectiveness of the Matheuristic Approach

In this section, we present a detailed computational study to test the effectiveness of our matheuristic. We run all experiments on a standard workstation (Apple Silicon M3, 2024) with 8 CPU cores and 24 GB of RAM. The implementation was developed in Python 3.12, using Gurobi 12.0.2 as a MILP solver. Throughout this section, and based on preliminary experiments, we set $\eta = 5$ customers and $\Delta = 50$ to define neighboring clusters when applying our merging procedure described in Section 4.2.

6.1. Test Instances

As discussed in Section 2, our work is the first to address a tactical fleet size and mix problem involving heterogeneous MCVs with flexible compartments. Therefore, to the best of our knowledge, there are no benchmark instances or solution approaches in the existing literature. While Ostermeier and Hübner (2018) is the only work that considers fleet size and mix decisions with MCVs, a direct comparison is not possible because neither the proposed LNS, the test instances, nor the detailed per-instance results are accessible.

To address this limitation, we use the publicly available instances generated by Henke et al. (2015) and Henke et al. (2019) for their homogeneous MCVRP with flexible compartments, which are also among the works closest to ours. These instances can be categorized as $|N|\_|P|\_|M|\_s$, where $|N|$ denotes the number of customers, $|P|$ the number of product types, $|M|$ the number of compartments, and s a supply parameter (according to Henke et al. (2015), e.g., $s = 1$ means that each customer demands at least one product type and at most one third of the available product types). We consider instances with three product types, three compartments (i.e., each compartment is dedicated to one product type), and $s \in \{1, 2, 3\}$. For the instances of Henke et al. (2015), this results in a total of 150 instances with 10 customers and three instances with 50 customers. From Henke et al. (2019), a total of 45 instances are available, five for each number of customers $|N| \in \{10, 15, ..., 45, 50\}$.

To evaluate the effectiveness of our matheuristic, we compare our results with those of Henke et al. (2015) and Henke et al. (2019), which are also publicly available in their manuscript. This comparison is conducted under the same problem setting considered in these works, which corresponds to a restricted version of the problem studied here. To further validate our matheuristic under the full problem described in Section 3, we also provide a comparison on small instances for which all feasible customer clusters can be exhaustively enumerated, allowing us to solve the complete version of Model (3) to optimality. The following sections present both comparisons.

6.2. *Comparison with Henke et al. (2015) and Henke et al. (2019)*

To ensure a fair comparison with Henke et al. (2015) and Henke et al. (2019), we adopt the following considerations, recognizing that we address a tactical fleet size and mix problem with heterogeneous vehicles rather than an operational routing problem with homogeneous vehicles. First, since the cost structure of a fleet size and mix problem is typically driven by fixed vehicle costs, which are not considered by Henke et al. (2015) and Henke et al. (2019), we focus on comparing the number of vehicles to avoid misleading conclusions from incomparable cost structures. Note that the number of vehicles is an input parameter for the MILP model and solution approaches proposed by Henke et al. (2015) and Henke et al. (2019), which they computed by solving a bin packing problem when generating their instances. Second, we only consider homogeneous MCVs (i.e., a single vehicle type) when generating vehicle configurations (see Section 4.1). Third, since Henke et al. (2015) and Henke et al. (2019) do not impose route duration constraints, we do not enforce these constraints when generating feasible customer clusters (see Section 4.2). Lastly, in line with Henke et al. (2015) and Henke et al. (2019), we consider a multi-stop delivery policy when solving Model (3) (see Section 4.3).

Table 4 summarizes the comparison between the number of vehicles obtained by our matheuristic and those achieved by Henke et al. (2015) and Henke et al. (2019) through their bin packing approach (we refer the reader to our GitHub repository for detailed results). The columns in the table report the total instances per category, the number of instances where our matheuristic achieved the same number of vehicles as Henke et al. (2015) and Henke et al. (2019), the number where our matheuristic found fewer vehicles, and the average run time (in seconds) per instance category.

Table 4: Comparison between the number of vehicles obtained by our matheuristic and those of Henke et al. (2015) and Henke et al. (2019).

Benchmark	Category	Instances	# Same vehicles	# Fewer vehicles	Run time (s)
Henke et al. (2015)	10_3_3_1	50	24	26	5.0
	10_3_3_2	50	28	22	5.0
	10_3_3_3	50	35	15	5.1
	50_3_3_1	1	1	0	7.3
	50_3_3_2	1	1	0	8.1
	50_3_3_3	1	1	0	8.4
Henke et al. (2019)	10_3_3_3	5	5	0	5.2
	15_3_3_3	5	5	0	5.3
	20_3_3_3	5	5	0	5.3
	25_3_3_3	5	5	0	5.5
	30_3_3_3	5	5	0	5.4
	35_3_3_3	5	5	0	5.7
	40_3_3_3	5	5	0	5.8
	45_3_3_3	5	5	0	5.8
	50_3_3_3	5	5	0	6.5

Table 4 shows the effectiveness of our matheuristic in reducing or maintaining the number of vehicles needed compared to the number of vehicles found by Henke et al. (2015). For instances with 10 customers, our matheuristic finds solutions with exactly one fewer vehicle in 52%, 44%, and 30% of the instances with supply parameters s equals to 1, 2, and 3, respectively. For larger instances with 50 customers, our matheuristic finds the same number of vehicles as Henke et al. (2015). When compared with Henke et al. (2019), Table 4 reports that our matheuristic consistently finds the same number of vehicles, regardless of the problem size. Note that, regardless of the instance category, our matheuristic (from generating vehicle configurations to returning the best solution found; see Figure 3) takes less than nine seconds on average.

6.3. Comparison with Optimal Solutions of Model (3) under the Full Problem Setting

To evaluate the effectiveness of our matheuristic under the full problem setting considered in this work, we now present a comparison on small 10-customer instances for which all feasible customer clusters can be exhaustively enumerated, which allows us to solve the complete version of Model (3) to optimality. To this end, we extend the instances of Henke et al. (2015) by introducing heterogeneous vehicles, fixed vehicle costs, and a maximum route duration constraint. Two vehicle types are considered: Type A with a capacity of 1,000 load units (as defined by Henke et al., 2015) and Type B with a capacity of 500 load units. The fixed cost of using one vehicle is 100 cost units for Type A and 60 cost units for Type B. These values are consistent with the real-world parameters used in our case study (Section 7), where larger vehicles are cheaper per unit of capacity.

Customer locations and demands are taken from Henke et al. (2015), while all other parameters follow our case study setting, including a variable cost of 10 per operating hour, an average speed of 30 km/h, a service time of 25 minutes per customer, and a compartment setup cost of 10 per additional compartment (see Section 7 for further details). We impose a maximum route duration of 10 hours. This value was derived by solving, for each instance in Henke et al. (2015), a Traveling Salesman Problem (TSP) visiting all customers with a single uncapacitated vehicle. Across all 150 instances tested, tour durations range from 7.93 to 11.36 hours (mean: 9.89 hours). In 59 instances (39%), the tour duration already exceeds 10 hours, making a single-vehicle solution infeasible due to the maximum route duration constraint alone. In the remaining cases, capacity constraints still require multiple vehicles, meaning one or both constraints are binding in each instance.

Table 5 reports the average performance metrics of the matheuristic relative to the complete version of Model (3) (i.e., the version including all feasible customer clusters) for each instance category in Henke et al. (2015), including the speedup factor, optimality cost gap, number of optimal solutions found (defined as solutions within a 0.01% optimality gap), number of solutions using the same vehicles, and number of solutions using the same fleet mix. For both the matheuristic and Model (3), vehicle routes are explicitly computed to evaluate variable transportation costs and to verify that maximum route duration constraints are satisfied (see Section 4.2.1 for further details).

Overall, the matheuristic finds 78 of 150 optimal solutions, with an average cost gap of only 1.67%. Importantly, the matheuristic accurately prescribes the key tactical decisions, matching Model (3) in the number of vehicles across all 150 instances and finding the same fleet mix in 139 of them. This performance is achieved while considering far fewer customer clusters: the matheuristic examines an average of only 79 customer clusters, compared to 1,409 clusters in the complete model (recall that only feasible customer clusters are considered). This translates to an average speedup factor of 7.75 over Model (3), with the matheuristic achieving an average run time of 33.3 seconds. Before concluding this section, it is worth noting that when defining vehicle routes implicitly through CA, the matheuristic also matches the number of vehicles in all 150 instances and the same fleet mix in 139 instances. This indicates that both the implicit- and explicit-routing versions of the matheuristic produce the optimal number of vehicles and, in nearly all cases, identical fleet compositions, which constitute the main tactical decisions of interest.

Table 5: Average performance metrics of the matheuristic relative to the complete version of Model (3), considering Henke et al. (2015)'s instances with 10 customers.

Category	Instances	Speedup factor	Gap (%)	# Opt. solutions	# Same vehicles	# Same mix
10_3_3_1	50	7.50	1.35	29	50	47
10_3_3_2	50	8.06	1.20	27	50	49
10_3_3_3	50	7.70	2.47	22	50	43

7. Real-World Case Study

After showing the computational performance of our matheuristic approach, we now present and discuss results from a real-world case study based on the current operations of our industry partner, a leading food distributor based in Bogotá, Colombia. We begin this section by describing our real-world problem instances. We then present and discuss managerial insights regarding the benefits of adopting MCVs. Lastly, we analyze the resulting fleet composition under different cost structures.

Throughout this section, we employ the same parameters and computational environment described in Section 6. For transparency, detailed computational results (including run times, which are under 1 minute across all instances and parameter combinations tested) are publicly available at `https://github.com/ekohan/fleetmix`.

7.1. Real-World Problem Instances

Our industry partner provided us with a rich dataset containing 70 days of real-world demand information. Each day (i.e., problem instance) provides detailed information about customer locations and the quantities of products requested by product type (frozen, chilled, and dry). Over the 70 days, the number of customers served by the company ranges from 208 to 691, with an average of 379 customers. Below, we explain the main characteristics of the service area and vehicle fleet, as well as the parameter values used.

Service area. The service area is located in Bogotá, Colombia, the country's most populous city. Geographically, the service area spans around 1,226 km^2. The company serves an average of 379 customers per day from a single depot located on the outskirts of the city. Figure 4 shows the spatial distribution of customers for a representative day (to protect confidentiality, the node locations have been slightly perturbed).

Heterogeneous MCVs. The company operates a fleet of heterogeneous MCVs, comprising three vehicle types with flexible compartments. Before the delivery horizon begins, the company decides the number of each vehicle type to deploy and their respective compartment configurations. Vehicle Type A has a total capacity of 700 kg and a fixed cost of 80 units; Vehicle Type B has a capacity of 1,300 kg and a fixed cost of 140 units; and Vehicle Type C provides 2,500 kg at a fixed cost of 180 units (costs have been scaled for confidentiality). Each vehicle type allows seven compartment configurations (see Section 4.1), with a (homogeneous) setup cost of 10 units per associated compartment. Thus, for example, the total fixed cost of using one vehicle of Type C equipped with two compartments (e.g., dry and frozen) is 200 units (i.e., $180 + 2 \cdot 10$). We assume that vehicles travel the Euclidean distance between nodes at a constant speed of 30 km/h. Based on the company's recommendation, we set a service time of 25 minutes per stop, which is the average time the company currently takes to deliver food products to their customers (e.g., local restaurants or hotels). The variable vehicle

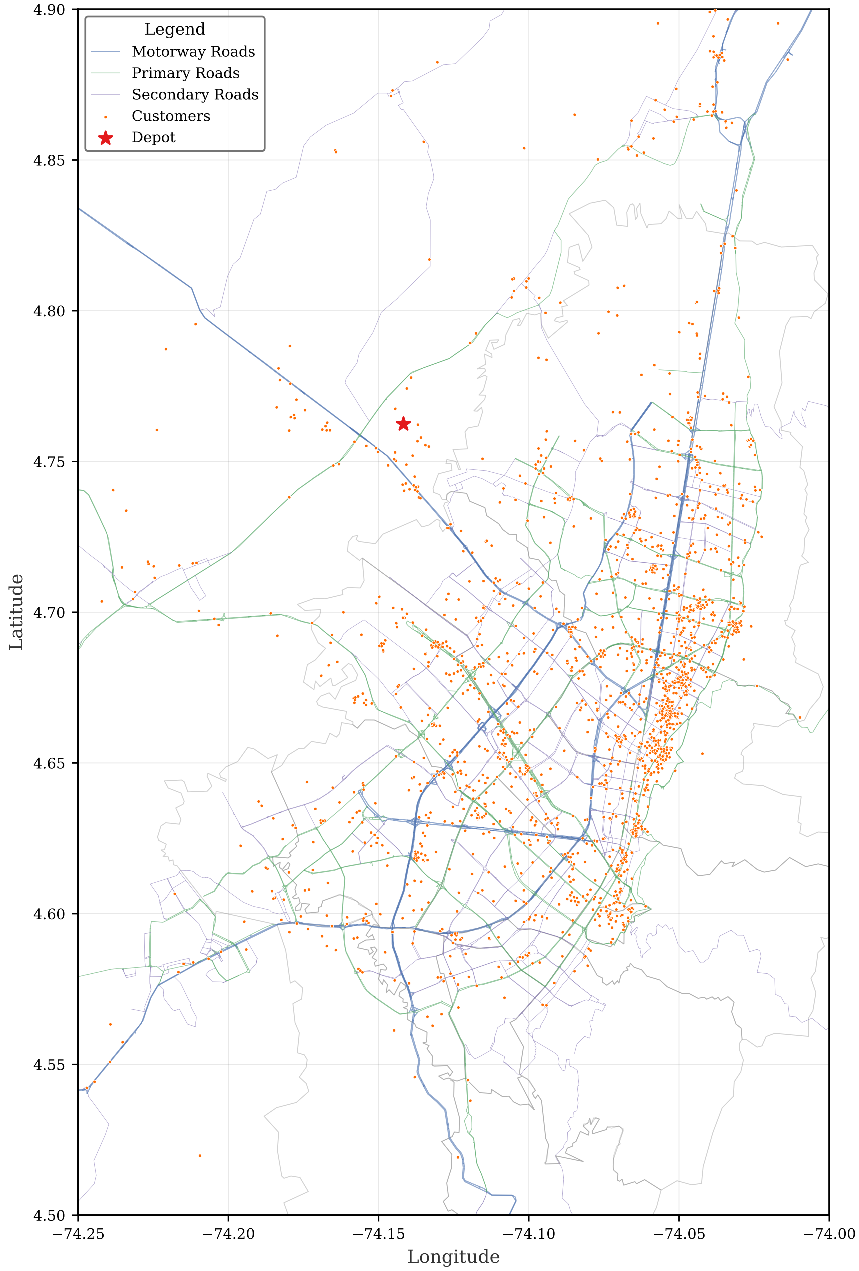


Figure 4: Service area and customer distribution.

cost is 10 units per operating hour, with a maximum route duration of 10 hours. Lastly, given that our instances involve hundreds of customers (from 208 to 691 customers), we check route duration feasibility through CA when generating customer clusters, using a BHH constant of $\beta = 0.765$ (see Section 4.2.1 and Carlsson and Yu (2025) for a detailed discussion of this parameter).

7.2. Benefits of Using Multi-Compartment Vehicles

To understand the benefits of using MCV fleets over traditional SCV fleets, we assess the effects of varying three key operational parameters: total vehicle capacity, customer service time, and the maximum allowed route duration. We establish the baseline scenario with the parameter values detailed in Section 7.1, which capture the company's current operations. For each parameter variation, we compute the impact on fleet size (i.e., the total number of vehicles employed) and total cost (comprising fixed and variable costs associated with using the selected vehicle configurations).

Figure 5 illustrates how the average fleet size and total cost change when varying a single parameter by –50%, –20%, +20%, and +50% relative to the baseline scenario. Results consistently show that adopting MCVs leads to reductions in both metrics across all parameter combinations. Notably, in the baseline scenario (indicated by the vertical line at 0% on the horizontal axis of each subplot), using MCVs reduces the average fleet size by 9.0 vehicles and the average total cost by 18.5%. To facilitate the interpretation of these results, Table 6 reports additional operational metrics: vehicle load utilization (i.e., the average percentage of total vehicle capacity used), the average number of customers served per vehicle, the average route duration (including travel and service times), and the fleet composition (in terms of vehicle types used). We now examine the main direct effects of each parameter variation.

Table 6: Average operational metrics (over 70 instances) per parameter variation.

Parameter	Variation	Vehicle load utilization		Customers per vehicle		Route duration		MCV fleet composition		
		MCV	SCV	MCV	SCV	MCV	SCV	Type A	Type B	Type C
Load Capacity	-50%	87.5%	85.8%	10.1	12.2	5.3	6.6	6.2	3.0	16.9
	-20%	86.5%	79.6%	12.4	13.4	6.5	7.1	6.8	3.2	10.5
	0%	86.4%	75.7%	13.4	13.7	7.0	7.3	7.5	3.7	7.5
	20%	83.6%	71.8%	14.2	14.1	7.4	7.5	8.8	3.0	5.8
	50%	81.7%	66.4%	14.6	14.4	7.6	7.6	10.9	2.6	3.7
Max. Route Duration	-50%	77.6%	57.1%	8.1	8.0	4.4	4.4	24.3	2.4	4.3
	-20%	84.9%	72.2%	11.6	11.8	6.1	6.3	11.7	3.2	6.6
	0%	86.4%	75.7%	13.4	13.7	7.0	7.3	7.5	3.7	7.5
	20%	86.4%	78.3%	15.1	15.3	7.8	8.1	5.2	3.2	8.4
	50%	85.6%	79.0%	16.3	17.6	8.5	9.3	3.9	2.3	9.3
Service Time	-50%	85.6%	78.5%	17.3	19.2	5.2	5.9	3.0	2.1	9.6
	-20%	86.0%	77.0%	15.0	15.4	6.5	6.9	5.4	2.9	8.5
	0%	86.4%	75.7%	13.4	13.7	7.0	7.3	7.5	3.7	7.5
	20%	85.7%	73.6%	12.2	12.4	7.4	7.7	10.4	3.3	6.9
	50%	84.9%	70.3%	10.7	11.0	7.8	8.1	14.4	2.8	6.1

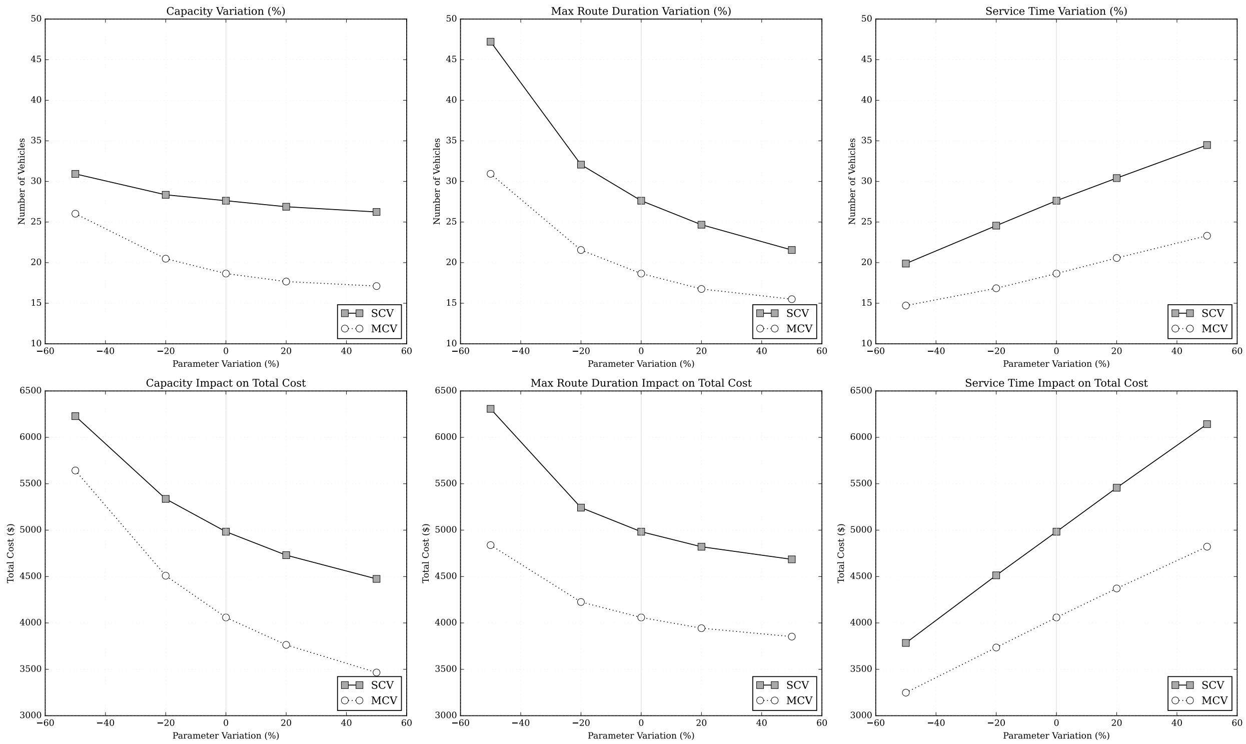


Figure 5: Average fleet size and total cost (over 70 instances) achieved by the MCV and SCV fleet for each parameter combination.

Effects of vehicle capacity. Figure 5 shows that vehicle capacity has only a modest effect on fleet size for both MCV and SCV cases. For instance, a 50% reduction in the total vehicle capacity results in only a slight increase in the number of vehicles (approximately from 18.7 to 26.0 for MCVs and from 27.6 to 30.9 for SCVs) yet leads to a sharp rise in total cost. Table 6 supports these findings: when load capacity is reduced by 50%, the MCV fleet composition shifts toward a much higher share of large, more expensive Type C vehicles (from 7.5 to 16.9 vehicles on average), while the number of small, less expensive Type A vehicles decreases. This substitution keeps the total number of vehicles required almost unchanged because larger vehicles can absorb the reduced capacity, but it substantially increases total costs. These results show that changes in capacity mainly affect total cost rather than fleet size, since the fleet composition can be adjusted without substantially changing the number of vehicles required.

Effects of maximum route duration. As shown in Figure 5, the response to changes in maximum route duration is highly asymmetric. For instance, when the maximum route duration is reduced by 50%, the number of vehicles increases by approximately 66% and 71% in MCV and SCV cases, respectively, resulting in a substantial increase in total cost: 19% for MCVs and 27% for SCVs. Conversely, extending the maximum route duration by 50% (corresponding to 15 working hours, which exceeds practical operational limits but is included here for illustrative purposes) yields only moderate efficiency gains: the average fleet size decreases by 17% and 22% for the MCV and SCV cases, respectively, with total cost reductions of only 5.1% and 6.0%. This asymmetric response is mainly driven by MCV fleet composition: with a 50% reduction in maximum route duration, small vehicles (Type A) rise quickly from 7.5 to 24.3, since limited time on route prevents efficient use of (more expensive) larger vehicles. In contrast, when duration is extended by 50%, the share of large vehicles (Type C) increases from 7.5 to 9.3, but their higher capacity does not translate into a substantial reduction in total fleet size compared to the baseline scenario (e.g., when using MCVs, the total fleet size is reduced from 18.7 to 15.5 vehicles). Overall, the results suggest that the company should not focus on extending maximum route durations (i.e., drivers' working hours) to improve performance, as this provides only limited benefits.

Effects of service time. Figure 5 shows that customer service time exhibits the most linear relationship among the parameters studied. If the company manages to reduce service time by 50% (e.g., through better coordination with customers regarding product unloading and delivery), the average number of MCVs can be reduced by 3.9 vehicles, while achieving a reduction of 7.7 vehicles when employing a traditional SCV fleet. This pattern is also reflected in Table 6. As service time decreases, vehicles can serve more customers per vehicle in a nearly proportional way (from 13.4 to 17.3 customers for MCVs and from 13.7 to 19.2 for SCVs under a 50% reduction). Similarly, the effective route duration follows the same trend, decreasing almost linearly as service time is reduced (from 7.0 to 5.2 hours for MCVs and from 7.3 to 5.9 for SCVs). Because both customers-per-vehicle and route duration scale almost directly with the service time parameter, the total number of required vehicles decreases at a nearly constant rate as service time reduces. Therefore, this nearly linear relationship means that any effort to reduce service time results in a proportional reduction in both fleet size and total costs.

7.3. Impact of Cost Structure on Fleet Composition

Due to the practical challenges associated with managing multiple vehicle types, our industry partner is currently considering the use of a homogeneous vehicle fleet (composed of a single vehicle type). Accordingly, in this section, we examine how the cost structure associated with using MCVs influences fleet composition decisions. For this analysis, we consider that the vehicles in the fleet have a total load capacity of 2,700 kg and can be designated either as SCVs or MCVs with flexible compartment sizes.

For this analysis, we establish a baseline fixed cost of 100 units for using one SCV. The fixed cost for using one MCV with configuration $v \in V$ is set as $100 \cdot (\alpha + c \cdot (|M_v| - 1))$, where $\alpha \in [1, 2]$ represents the premium factor associated with using MCVs; M_v, with $\emptyset \neq M_v \subseteq M$, is the subset of compartments of vehicle configuration $v \in V$; and $c \in [0, 0.5]$ denotes a fraction of the SCV fixed costs incurred per additional compartment. For example, when $\alpha = 1.5$ and $c = 0.2$, the total fixed cost of using one MCV with three compartments is $190 = 100 \cdot (1.5 + 0.2 \cdot (3 - 1))$, as there are two additional compartments. All other operational parameter values (e.g., customer service time, variable costs) remain the same as defined in Section 7.1.

Figure 6 presents the combined effect of the premium factor (α) and compartment setup factor (c) on fleet composition. The horizontal axis represents the compartment setup factor, which increases from 0% to 50% in increments of 10%. The vertical axis corresponds to the α multiplier, which scales the fixed cost of using one MCV from 1.0 to 2.0 in increments of 0.1. Each cell in Figure 6 displays three values: the average MCV share in percentage, the number of days (out of 70) in which at least one MCV is used, and the number of days in which all vehicles in the fleet are MCVs, marked with a star.

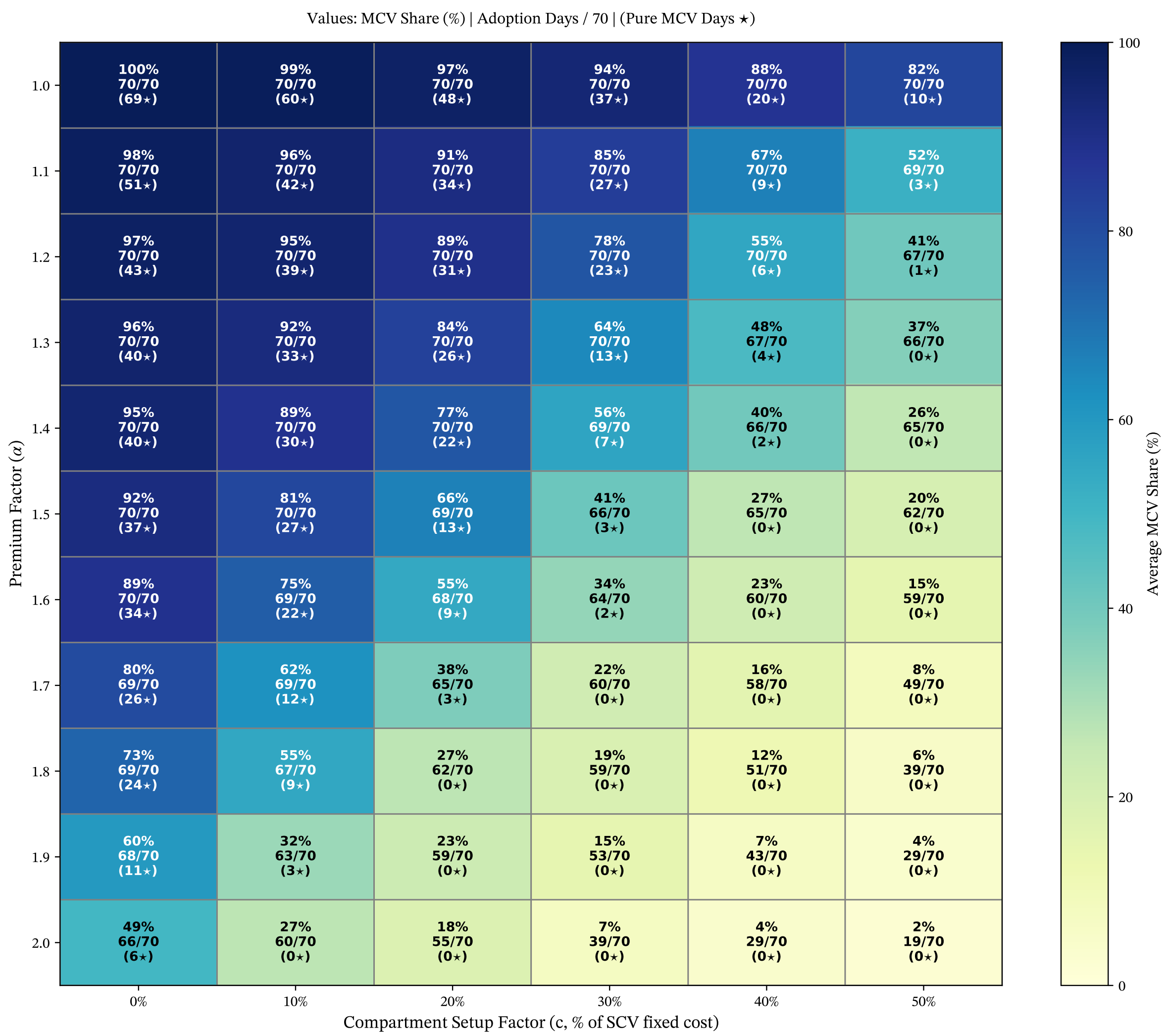


Figure 6: Effect of premium factor (α) and compartment setup factor (c) on fleet composition.

Figure 6 shows a clear nonlinear decline in MCV adoption as both cost factors increase. When MCVs are relatively inexpensive compared to SCVs (e.g., when $\alpha \leq 1.4$ and $c \leq 0.1$, or $\alpha \leq 1.2$ and $c \leq 0.2$), their share consistently represents 90% or more, indicating that MCVs dominate the fleet composition and achieve full adoption across all days. However, even modest increases in either parameter rapidly reduce their utilization. When $\alpha \geq 1.5$ and $c \geq 0.3$, the average share of MCVs falls below 50%, suggesting that their fixed cost quickly outweighs the operational flexibility offered by compartmentalization. Overall, Figure 6 shows that MCV adoption is highly sensitive to the underlying cost structure, indicating that only specific cost parameters make MCVs an operationally dominant option for fleet composition.

Lastly, Table 7 provides a detailed view of how the MCV share varies across different compartment setup cost factors, with the premium factor set to $\alpha = 1.6$. As the compartment setup factor increases from $c = 0$ to $c = 0.5$ (i.e., the cost of setting up an additional compartment in a MCV represents 50% of the cost of using a SCV), the average MCV share rapidly falls from 89% to 15%. Initially, this reduction in MCV usage might suggest a decrease in total costs, yet the opposite effect occurs. SCV fleets require more vehicles to handle incompatible product types separately, leading to an increase in the total fleet size. Consequently, customers are served by multiple vehicles more often. This increase in fleet size raises the total cost by approximately 15% (from 3,901 to 4,498 cost units), as shown in Table 7. In summary, these results confirm that compartment setup costs have a strong effect on the daily fleet composition, consistent with the overall pattern observed in Figure 6.

Table 7: Impacts of cost structure when $\alpha = 1.6$. Results averaged over 70 instances.

Compartment setup factor (c)	MCV share	Total cost	Fleet size	Visits per customer
0.0	89%	3,901	16.9	1.1
0.2	55%	4,292	19.0	1.3
0.5	15%	4,498	23.0	1.5

7.4. Managerial Insights

The results of our real-world case study provide valuable insights for food distributors and fleet managers considering the adoption of MCVs. Below, we summarize the key managerial implications derived from our analysis.

Multi-compartment vehicles reduce both total costs and fleet size. Our results show that adopting MCVs can lead to substantial performance gains compared with traditional last-mile fleets based on SCVs. Under the current operating conditions of our industry partner (i.e., our baseline scenario), we find that total costs decrease by 18.5% and that the total number of required vehicles decreases by an average of 9.0. This improvement comes from the ability of MCVs to consolidate incompatible product types, allowing vehicles to depart with fuller loads, make fewer stops, and spend less time distributing.

Prioritize reducing service time to boost performance. Our sensitivity analysis shows that service time is the most practical and effective lever for improving performance, since even small reductions produce clear gains in fleet size and total cost. In contrast, increasing vehicle capacity or extending working hours (i.e., maximum route duration limits) requires additional investment and offers only limited benefits. Changes in capacity have minimal impact on the number of vehicles required, and longer route durations yield only modest improvements. Because shorter customer service times translate almost directly into operational efficiencies, efforts should focus on streamlining delivery and unloading processes (e.g., by improving coordination with customers) rather than investing in larger vehicles or extending working hours.

Deploy multi-compartment vehicles when cost structure supports their flexibility. Our findings show that fleet size and mix decisions with MCVs are sensitive to their underlying cost structure. MCVs are only an operationally dominant option when their fixed-cost structure stays within a favorable range; once premiums or compartment setup costs increase, the fleet composition shifts toward SCVs. This shift forces more vehicles onto the road and increases how often customers are visited by multiple vehicles, thereby increasing total costs.

8. Conclusions

Multi-Compartment Vehicles (MCVs) are among the most effective vehicle technologies that food distributors can employ to consolidate deliveries of multiple product types while maintaining their specific temperature and loading conditions. Motivated by our collaboration with a major food distributor operating in Bogotá, Colombia, we investigate the design of last-mile delivery fleets involving heterogeneous MCVs with flexible compartment sizes. Specifically, the decision-making problem faced by the company considers three interconnected decisions: (i) the number of each vehicle type to use, (ii) the type and size of each compartment assigned to each vehicle, and (iii) the distribution plan for each vehicle. The objective is to minimize total logistics costs, which comprise both fixed and variable costs associated with selecting and utilizing vehicles.

To solve this problem, we propose a novel matheuristic approach based on a cluster-first, fleet design-second scheme. Our matheuristic can accommodate several real-world characteristics of modern food distribution operations, including homogeneous and heterogeneous vehicles with fixed or flexible compartment sizes; a comprehensive cost structure that allows including fixed costs, compartment-dependent costs, and variable (distance or time-dependent) costs; and single- and multiple-stop delivery policies. To facilitate the application of our matheuristic, we create FleetMix, an open-source Python package (MIT license, PyPI). Its implementation follows specifications linking the paper's methodology to its software realization, uses defined interfaces to allow for systematic adaptation of its components, and includes a benchmarking tool to ensure verifiable replication of our results.

Using established synthetic instances from the literature, we show that FleetMix achieves high-quality fleet designs compared to existing solution approaches. Through a real-world case study informed by real data from our partner industry company, we derive several managerial insights into the value of using MCVs. Our results show that, due to their ability to consolidate multiple product types, MCVs can substantially reduce total logistics costs and fleet size by enabling vehicles to depart with fuller loads and perform more efficient routes. We also find that reducing customer service time is the most effective lever for improving performance compared with increasing vehicle capacity or extending operating hours.

Since our work is among the first to address the tactical design of last-mile delivery fleets with MCVs, there are several opportunities for future work. First, while this is a challenging optimization problem, future research could focus on developing exact solution approaches that can solve practical problem instances to proven optimality. Here, one could develop a column generation-based scheme and use our set partitioning formulation presented in Section 4. Second, future work can also extend FleetMix to accommodate other real-world characteristics, such as other cost components (e.g., loading and unloading costs, as in Hübner and Ostermeier (2019)) and time windows. Lastly, a stochastic version of the problem, where tactical fleet size and mix decisions are made before customer demand is known, will be a challenge to address.

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Appendix A. Clustering Techniques

To create customer clusters, we use a variety of methods, including k-means, k-medoids, Gaussian mixture, and hierarchical clustering (agglomerative). These methods can be used individually or in combination through what we call the *combine* method. Table A.8 summarizes and compares these clustering techniques.

Table A.8: Comparison of Clustering Methods

Method	Weight	Parameters	Advantages
k-means	$\lambda = 1$	Number of clusters	Fast convergence, simple to implement
k-medoids	$\lambda = 1$	Number of clusters, medoid selection	Robust to outliers
Agglomerative	$\lambda \in [0, 1]$	Linkage criteria and λ weight	Captures both spatial and demand factors
Gaussian Mixture	$\lambda = 1$	Number of clusters, covariance type	Flexible cluster shapes, probabilistic cluster assignment
Combine	$\lambda \in [0, 1]$	All above parameters	Ensemble approach, robust solutions, diverse clustering patterns